

Project Design Phase-II

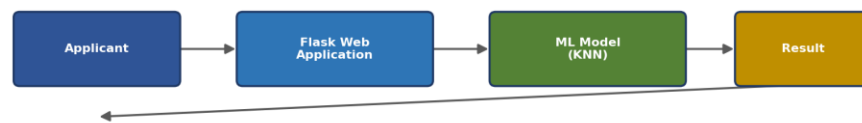
Data Flow Diagram & User Stories

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

About Data Flow Diagrams

A Data Flow Diagram (DFD) is a visual representation of how data enters and leaves a system, what transforms it, and where it is stored. The diagrams below trace applicant data from input through to the final loan decision.

Level 0 - Context Diagram



Level 1 - Detailed Flow

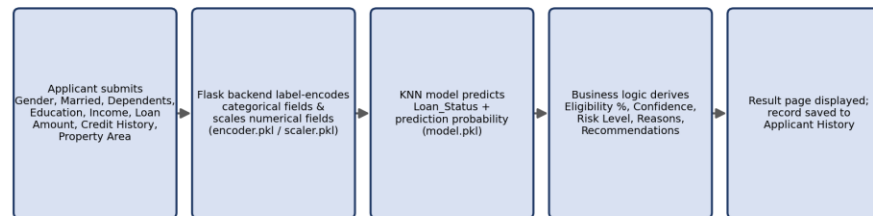


Fig 1: Data Flow Diagram – Level 0 & Level 1

Level 0 – Context Diagram

- External Entities: Applicant, Flask Web Application, ML Model (KNN).
- Process: “Loan Approval Prediction.”
- Data Stores: model.pkl, encoder.pkl, scaler.pkl, prediction history.
- Data Flow: Applicant → Loan Application Form → Flask App → ML Model → Prediction → Result Page → Applicant.

Level 1 – Detailed Flow

- Applicant submits Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History and Property_Area on the Predict page.
- Flask backend label-encodes categorical fields and scales numerical fields (scaler is applied for the KNN model).
- The KNN model predicts Loan_Status (Approved / Rejected) and returns a prediction probability.

- Application logic converts the probability into eligibility percentage, confidence score, risk level, reasons and recommendations.
- The Result page displays the outcome; the record is stored and shown later on the History page.

User Stories

User Type	Functional Requirement	User Story #	User Story / Task	Acceptance Criteria	Priority
Applicant	Loan Application	USN-1	As an applicant, I can fill and submit the loan application form with my personal and financial details.	Form validates required fields and submits successfully.	High
Applicant	Prediction & Decision	USN-2	As an applicant, I want the system to predict whether my loan will be approved or rejected.	Result page shows Approved/Rejected with eligibility %.	High
Applicant	Result Understanding	USN-3	As an applicant, I want to see the reasons and recommendations behind my result.	Result page lists reasons and next-step recommendations.	Medium
Applicant	History Tracking	USN-4	As an applicant, I can view my previous prediction records.	History page lists past predictions with details.	Medium
Loan Officer	Applicant Profile	USN-5	As a loan officer, I can view an applicant's profile details.	Applicant Profile page displays submitted information.	Low